Problem Set 1: Scale, Units, and Microscopy

Due: Sunday, August 30 at 11:59 PM **Value: 25 points** **Format: Five-question problem set**

Purpose

Practice the dimensional reasoning required to move between microscopic measurements, field of view, magnification, and image scale.

Your task

- Q1 (5): Convert 0.084 mm to micrometers and nanometers. Show conversion factors.
- Q2 (5): A 2.0 mm stage-micrometer span covers 80 ocular divisions. Determine micrometers per division.
- Q3 (5): A cell spans 7.5 ocular divisions using the calibration from Q2. Estimate cell length with appropriate significant figures.
- Q4 (5): A circular field of view is 450 micrometers across. Estimate how many 32-micrometer cells fit end to end across the diameter and explain why this is not the number visible in the full area.
- Q5 (5): A printed micrograph is 96 mm wide and represents 240 micrometers. Draw and label a 50-micrometer scale bar and calculate its printed length.

Required sources or materials

- BIO 210 unit-conversion guide
- Microscopy calibration example from Laboratory Manual, pp. 4-6

Submit

- One PDF showing equations, units, answers, and brief interpretations.

Grading criteria

Criterion	Pts	Full-credit evidence
Question 1	5	Correct setup, units, and biological interpretation.
Question 2	5	Correct reasoning shown; not just a final answer.
Question 3	5	Mechanism-based explanation uses appropriate evidence.
Question 4	5	Prediction follows from the stated model and conditions.
Question 5	5	Data are represented or evaluated accurately.

Submission standards

Upload a readable PDF unless the handout specifies another format. Include your name, course code, and assignment title. Cite borrowed ideas and retain the editable source file. The learning portal timestamp is the official submission time.